

Master Project Proposal

KPI-RAG: Explainable Root-Cause Analysis for 5G Networks via Standards-Grounded Retrieval-Augmented Generation

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1 Abstract

Modern 5G networks generate continuous streams of Key Performance Indicator (KPI) data, and diagnosing service degradation currently requires network engineers to manually correlate raw metrics against technical standards, a slow, error-prone process that does not scale. This project proposes KPI-RAG, a three-layer pipeline that detects network anomalies from real 5G KPI signals, classifies their root cause across eleven fault categories, and generates standards-grounded natural-language explanations by retrieving relevant 3GPP and O-RAN specification content. The system trains and evaluates on TelecomTS, a 32,000-sample 5G testbed dataset combining real testbed recordings with synthetically injected faults, using two cascaded Random Forest classifiers (XGBoost as contingency) with SHAP-based explainability for detection and a novel Fault-to-Standard Alignment Table, cross-referenced against GSMA telecom-kg and oran-kg knowledge graphs and confirmed against primary specification documents, to ground LLM-generated explanations in verifiable technical citations. To the best of our knowledge, based on searches of IEEE Xplore, ACM Digital Library, and arXiv conducted in July 2026, neither TelecomTS nor TN-AutoRCA links a KPI-level anomaly taxonomy to formal 3GPP specification references in a validated, evaluable artifact; constructing and evaluating this mapping is this project's primary contribution. Anomaly detection and standards-compliant diagnostic reasoning are typically treated as separate, disconnected problems in AIOps research. This project addresses that gap directly, combining interpretable signal classification with standards-grounded explanation in a single evaluable pipeline.

2 Introduction / Background

5G Network Operations Centers (NOCs) must diagnose faults, signal degradation, interference, and congestion across increasingly complex infrastructure. Engineers currently correlate raw KPI measurements against technical specifications manually: slow, inconsistent, and unscalable with fault volume [6]. Without automation, every incident depends on individual expertise, prolonging mean-time-to-resolution (MTTR) and increasing operational cost. Telecom operators, NOC engineers, and end users all benefit from faster, consistent root-cause identification; the research community benefits from the first publicly documented mapping between a 5G anomaly taxonomy and formal 3GPP standards. Current diagnostic practices rely heavily on engineers manually interpreting KPI measurements and consulting technical documentation. The reasoning behind these decisions is rarely captured in a structured, reusable form, making diagnosis dependent on individual expertise and difficult to reproduce consistently across similar incidents.

KPI-RAG addresses this gap by connecting KPI-signal-level detection directly to a validated, citation-backed explanation layer within one evaluable pipeline, rather than treating anomaly detection and standards-compliant explanation as separate problems.

3 Problem Statement

Given a window of 5G KPI measurements, this project addresses three connected sub-problems: (1) detecting whether observations represent anomalous behavior; (2) classifying the fault type among eleven categories (ten synthetically induced, one real jamming event); (3) generating a root-cause explanation grounded in official 3GPP and O-RAN specifications. To the best of our knowledge, no published system addresses all three in a single evaluable pipeline on public data.

Scope boundaries: Single-cell, single-modality (KPI time-series) fault diagnosis on TelecomTS. Multi-cell correlation, cross-domain RCA spanning RAN, core, and transport, and live production deployment are out of scope and noted as future work.

Scope of Validity: KPI-RAG is evaluated on a single indoor 5G testbed with 10 synthetically injected and 1 controlled real anomaly type. Generalization to multi-vendor, multi-cell, multi-operator production deployments is explicitly out of scope. Production deployment would require re-evaluation on operator data with natively occurring fault labels.

Research Questions:

- **RQ1:** Does scale-preserving feature engineering (582-dim) improve anomaly detection F1 and fault classification F1-macro vs. normalized-feature and statistics-only baselines?
- **RQ2:** Does alignment-table grounding produce measurably higher citation faithfulness vs. ticket-only and ungrounded LLM generation?
- **RQ3:** Which feature configuration (full 582-dim, 326-dim without patchwise scale, or 64-dim statistics-only) offers the best performance-to-cost tradeoff?

4 Research Objectives

Primary Objective: Design, implement, and evaluate an end-to-end pipeline that detects and explains 5G network faults with verifiable standards grounding.

- **O1 — Anomaly Detection:** Train a scale-aware two-stage Random Forest pipeline on TelecomTS across 11 fault types from 18 KPI channels, quantifying the contribution of absolute-scale feature preservation. *Model 1:* binary F1 comparable to Mantis (F1=0.800, TelecomTS Table 2). *Model 2:* per-class F1-macro across all 11 classes; plain accuracy alongside for comparability with the TelecomTS RCA benchmark (best published: Toto accuracy 0.848, Mantis accuracy 0.590, Table 4).

- **O2 — Fault-to-Standard Alignment:** Construct and validate a 10-row Alignment Table linking synthetic fault categories to specific 3GPP clauses and O-RAN components (jamming excluded, no corresponding clause), cross-referenced against GSMA KGs and confirmed against primary specification text. *Criterion:* all 10 rows validated by two team members, zero unresolved disputes.
- **O3 — Standards Retrieval:** RAG pipeline using SHAP attributions as the structured retrieval query. *Criterion:* acceptable precision@k on held-out alignment-table queries, evaluated at Week 9.
- **O4 — Grounded Explanation:** Plain-language reports where every claim is grounded in a retrieved clause or ticket. *Criterion:* $\geq 70\%$ pass the regex-validated 3GPP clause check.
- **O5 — Evaluation:** Joint detection and explanation faithfulness protocol. No existing benchmark evaluates both jointly, making this a contribution in itself.

Summary of Contributions: (1) **Artifact:** first TelecomTS-to-3GPP/O-RAN alignment mapping with independently verified evidence spans for 10 fault types. (2) **Empirical finding:** Track C ablation measures whether standards grounding produces higher citation faithfulness than ticket-retrieval-only and ungrounded generation. (3) **Pipeline demonstrator:** first per-class F1 table for the 11-class TelecomTS task, paired with an evaluable end-to-end implementation.

5 Related Work

KPI-RAG fills the gap between anomaly detection and standards-grounded explanation, which no published system addresses in a single evaluable pipeline on public data. This review covers 11 directly cited works spanning four sub-areas, selected from a broader survey of the 2024–2026 AIOps and telecom-RCA literature.

5.1 Anomaly Detection in 5G Networks

- [1] Feofanov et al., “Mantis,” [arXiv:2502.15637, Feb. 2025](#). Strongest published result on TelecomTS (F1=0.800); fault labels only, no KPI attribution or standards reference.
- [2] Feng et al., “TelecomTS,” [arXiv:2510.06063, Oct. 2025](#). Introduces the 32,000-sample 5G testbed dataset used throughout this project.
- [3] Lin et al., “NuTime,” [TMLR 2024](#). Scale-preserving NME encoder; this project adopts the patchwise-statistics principle without the transformer backbone.
- [4] Malacarne et al., “Scalable Context-Aware Graph Attention,” [arXiv:2605.00482, 2026](#). Production KPI anomaly detection at Telenor; validates real-operator viability.

5.2 Telecom Knowledge Benchmarks and RAG Evaluation

- [5] Maatouk et al., “TeleQnA,” [IEEE Network, 2025](#). Demonstrates RAG improves LLM accuracy on telecom standards questions; used here as a secondary evaluation proxy only.

5.3 LLM-Based Root Cause Analysis

- [6] Zhang et al., “A Survey of AIOps,” [ACM Comput. Surv., 2025](#). Surveys 183 LLM4AIOps papers (2020–2024); confirms standards-grounded explanation evaluation remains underexplored.
- [7] Chen et al., “RCACopilot,” [EuroSys 2024](#). Production RAG-based RCA at Azure (micro F1=0.766); cloud metrics only, no 3GPP citations.
- [8] Roy et al., “LLM-Based RCA Agents,” [FSE 2024](#). Groundedness evaluation coding scheme adapted in this project.
- [9] Wu et al., “TN-AutoRCA,” [arXiv:2507.18190, Jul. 2025](#). First telecom alarm-based RCA benchmark; alarm/topology modality, dataset unavailable.
- [10] Sana et al., “Reasoning LLMs for 5G RCA,” [arXiv:2507.21974, Jul. 2025](#). Domain fine-tuning outperforms general-purpose models; drive-test modality differs.
- [11] Y. Liu, D. Iter, Y. Xu, S. Wang, R. Xu, and C. Zhu, “G-Eval: NLG Evaluation Using GPT-4 with Better Human Alignment,” in *Proc. EMNLP*, 2023. [arXiv:2303.16634](#). Provides the evaluation framework used to apply the four project-defined dimensions to generated explanations; peer-reviewed, widely adopted for LLM output evaluation.

5.4 Summary of Gaps

To the best of our knowledge, no existing system connects KPI-signal-level anomaly detection to specific 3GPP clause-cited root cause explanation in a single evaluable pipeline on a public benchmark dataset. No published work links TelecomTS fault categories to specific 3GPP clauses and O-RAN components. This project addresses both gaps simultaneously. Explanation quality is evaluated using G-Eval [11] with four project-defined dimensions appropriate for the telecom standards-citation context.

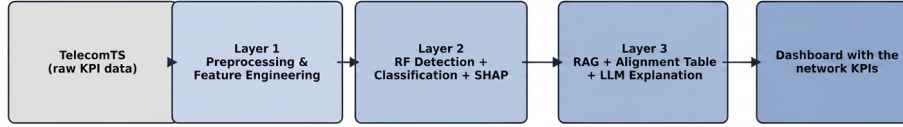
6 Proposed Approach / Methodology

6.1 High-Level Approach

KPI-RAG is a three-layer pipeline that takes a raw 5G KPI time series as input and produces an interactive, standards-grounded fault diagnosis dashboard as output. The three layers run sequentially during inference, culminating in a Streamlit dashboard deployed on HuggingFace Spaces that displays the detected fault type, an uncalibrated model confidence score, a SHAP bar chart of top contributing KPI channels, and a structured root-cause explanation grounded in a specific 3GPP clause and O-RAN component. When the binary detector outputs normal, the pipeline terminates at Layer 2 and the dashboard displays a “No anomaly detected” message; Layer 3 is not invoked.

6.2 System Architecture

Layer 1 — Preprocessing and Feature Engineering: loads TelecomTS at native scale; constructs the 582-dimensional feature vector per sample (64 precomputed statistics + 256 scale-preserving patchwise statistics + 256 first-order differences + 6 categorical encodings); sliding window length 128, stride 32; exports a stratified window-level train/test split ($\approx 80/20$, stratified by 12-class



fault label) as shared NumPy arrays for both models. Recording-level split was evaluated but found infeasible: all 1,235 anomalous windows are concentrated in 2 of 33 recording sessions, producing a test set with zero anomalous samples. The window-level split is consistent with the TelecomTS paper’s own evaluation protocol (Section 4.1).

Layer 2 — Anomaly Detection, Fault Classification, and SHAP: two cascaded Random Forest models (XGBoost as a contingency if F1 falls below the model-lock threshold at Week 5), a binary detector followed by an 11-class classifier, with SHAP TreeExplainer producing per-channel attribution scores. Output: structured JSON consumed by Layer 3.

Layer 3 — RAG and Standards-Grounded Explanation: retrieves top- k similar historical tickets from TelecomTS (ChromaDB), injects the pre-validated 3GPP clause and O-RAN component from the Alignment Table, and generates a structured explanation via an open-source LLM.

6.3 Technical Details

Layer 1. Scale is preserved throughout; no StandardScaler or MinMaxScaler is applied. Random Forest classifiers are invariant to monotonic feature transformations, but absolute-magnitude information must be preserved at the feature-extraction stage to be encoded in classifier inputs: normalizing the raw time-series before feature extraction discards this information. Mean RSRP = -100 dBm is diagnostically different from -70 dBm; this difference must survive into the 582-dimensional vector. TelecomTS benchmark results confirm scale-aware models outperform normalized-input models on this specific dataset.

Each 128-timestep window produces a 582-dimensional feature vector:

Component	Dims	How Computed	Why
Precomputed statistics	64 (4×16)	Re-computed from raw KPI values in the KPIs field: mean, std, min, max per channel. TelecomTS statistics, anomalies, labels, QnA, description, and troubleshooting_tickets fields explicitly excluded from classifier inputs	Directly interpretable per-channel summaries; computed from raw KPI values to prevent any label-field contamination
Patchwise scale stats	256 ($8 \times \text{mean} + \text{std} \times 16$)	128 timesteps split into 8 patches; mean and std per patch per channel	Preserves absolute magnitude within each patch; key for scale-aware detection
First-order differences	256 ($8 \times \text{mean} + \text{std} \times 16$)	Diff array; 8 patches $\times$ mean+std over the difference signal	Captures rate-of-change for temporal fault progression
Categorical encoding	6	One-hot UL/DL_Protocol (TCP/UDP/None)	Protocol state is diagnostic under fault
Total	582		

Patch count 8 was chosen to balance temporal resolution against dimensionality. The dataset is partitioned using a stratified window-level split (80/20), stratified by the 12-class multiclass label to guarantee all fault types are represented in the test set. Recording-level split was evaluated and rejected: anomalous windows are concentrated in 2 of 33 sessions, producing a test set with zero anomalous samples—making evaluation impossible. This window-level approach is consistent with the TelecomTS paper’s own protocol. The 75% overlap between adjacent train and test windows is acknowledged as a limitation. The same partition index (train_idx.npy, test_idx.npy) applies to the RAG corpus: only tickets from training-partition windows are indexed in ChromaDB. The two smallest classes (Congestion-Sudden $n = 48$, Resource-Bugs $n = 60$) use stratified 5-fold CV, reported as separate rows with mean $\pm$ 95% CI.

Layer 2. Model 1 (binary): RF trained on an 80/20 normal/anomalous rebalanced dataset, consistent with the TelecomTS paper’s own protocol; class_weight=‘balanced’ provides additional compensation; the test partition is never resampled (evaluated at natural $\approx 96\%/4\%$). Both models are tuned via RandomizedSearchCV with stratified k-fold on the training partition; if the model-lock thresholds are not met after tuning, further re-training stops and the shortfall is reported as a documented finding. An Isolation Forest runs in parallel as an unsupervised baseline. Model 2 (11-class): RF on anomalous-only samples (XGBoost contingency only). The cascade is evaluated via a full 12-class confusion matrix computed over all held-out samples, correctly accounting for Model 1 false positives and negatives. A conditional diagnostic metric (Model 1 recall $\times$ Model 2 accuracy on true anomalies) additionally isolates per-stage error sources. SHAP TreeExplainer yields top- K per-channel attributions with direction and magnitude in milliseconds (no GPU); these feed the Layer 3 query constructor and the dashboard.

Layer 3. *Component A (ticket retrieval):* The 1,235 TelecomTS troubleshooting tickets and associated QnA anomaly reasoning traces, one set per anomalous sample, from the troubleshooting_tickets nested field and the QnA.anomalies field of each anomalous JSONL record, are indexed in ChromaDB using all-MiniLM-L6-v2. Only content from anomalous samples (anomalies.exists = True) is indexed; normal-sample descriptions and QnA entries are excluded. The 1,235 count corresponds exactly to the 1,235 anomalous samples (confirmed by dataset inspection: 30,765 normal + 1,235 anomalous = 32,000 total).

For each anomaly event, a natural-language retrieval query is constructed from the Layer 2 JSON: *[PREDICTED_FAULT_TYPE] anomaly detected. Primary affected KPIs: [TOP_3_BY_SHAP]. Signal direction: [above/below the normal-class mean for each of the top-3 channels]. Protocol state: [UL/DL values]*. The SHAP sign determines direction: positive SHAP pushes toward anomalous; the direction label reflects whether the channel value lies above or below the normal-class mean in the training data. The template string is embedded and compared via cosine similarity; precision@3 and @5 are evaluated at the Week 9 checkpoint. If maximum similarity falls below the provisional threshold (0.45, to be calibrated at Week 9 via a precision-recall curve on held-out validation queries), the system flags low confidence and falls back to alignment-table-only generation with a “low historical precedent” dashboard warning.

Component B (standards grounding, the novelty): The Fault-to-Standard Alignment Table maps the 10 synthetic TelecomTS fault types to specific 3GPP Technical Specification clauses and O-RAN components. Built in Weeks 3 – 4, cross-referenced against GSMA telecom-kg-rel19 [D2] and oran_spec_knowledge_graph [D3] for reference discovery, with each row’s final citation confirmed against primary specification text (document ID, release, clause, and quoted evidence span). Jamming (11th class) shows “No 3GPP standard applicable” and uses ticket retrieval only.

Component C (LLM generation): Retrieved tickets and alignment row are fused into a structured prompt; the LLM returns a JSON object with five fields (root_cause, 3gpp_reference, oran_component, recommended_action, reasoning_trace). The 3gpp_reference field is validated by two checks: (1) format regex for series 21–38, three-digit sub-number; (2) cross-check against the alignment table’s known-valid TS reference list. Hallucinated-but-valid-format citations are flagged separately. JSON schema is enforced via outlines/guidance (local Llama) or JSON mode (Groq); two failed retries trigger a deterministic template-based fallback from the alignment row, flagged in results as template-generated.

Deployment: Queen’s University HPC cluster (Frontenac) for development and evaluation. Groq API (free tier, OpenAI-compatible) for the HuggingFace Spaces demo, subject to Capgemini data-governance confirmation before Week 7.

6.4 Design Decisions

Decision	Choice	Rationale
Classifier	RF (primary); XG-Boost as contingency if F1 falls below model-lock threshold	Native SHAP TreeExplainer compatibility (exact, millisecond-speed, CPU-only); Transformer+NME estimated at 5–6 weeks extra with SHAP/DeepExplainer friction; retained as future work
Dataset	TelecomTS only	Multi-dataset designs require a synthetic taxonomy adapter decoupling evaluation from demo; TelecomTS’s 11 labels serve directly as root causes
RAG source	Tickets + reasoning traces (primary) + static table (novelty)	Tickets and QnA traces are domain-specific, anomaly-annotated, and embedded-ready; live Neo4j/Cypher KG querying estimated at 3–4 weeks for zero inference-time gain over the static table
LLM	Interchangeable instruction-tuned LLM	Selected during implementation on reasoning quality, latency, hardware, licensing, and cost
Output	Streamlit + structured JSON	One view for signal+SHAP+explanation; JSON makes faithfulness a deterministic check
Novelty	Alignment Table only	Novelty concentrated in one testable, ablatability artifact; everything else uses proven techniques

6.5 Data

TelecomTS [D1]: 32,000 samples, 18 KPI channels (16 numerical, 2 categorical), 128 timesteps at 100ms, 11 anomaly types plus normal class. MIT license. Feng et al. (arXiv:2510.06063, 2025); HuggingFace. The 1,235 troubleshooting tickets and QnA anomaly reasoning traces within the dataset serve as the RAG retrieval corpus. **GSMA telecom-kg-rel19** [D2]: 3GPP Rel-19 KG; static alignment cross-reference only. CC BY-NC 4.0. **GSMA oran_spec_knowledge_graph** [D3]: O-RAN KG, 25,103 nodes, 98,679 relationships. CC BY-NC 4.0. **TeleQnA** [D4]: 10,000 telecom Q&A pairs; secondary evaluation proxy only. All datasets non-commercial academic use; no personal data; no human subjects.

6.6 Implementation Plan

The pipeline is implemented across ten modules assigned to four roles. P1 handles data preprocessing, feature engineering, and the scale ablation study. P2 owns both classifiers and the SHAP explainability layer. P3 builds the Fault-to-Standard Alignment Table and the RAG retrieval pipeline. P4 integrates the LLM explanation layer, runs the full evaluation protocol, and delivers the Streamlit dashboard. All modules communicate via a shared JSON schema, making each module independently testable before integration.

#	Module (Owner)	Description → Outputs
1	DatasetBuilder (P1)	Load TelecomTS; build 582-dim vectors; stratified window-level split (stratified by 12-class label; recording-level split evaluated and rejected—see Sec. 6.3) → NumPy train/test arrays; split indices saved as train_idx.npy and test_idx.npy; feature matrix documented
2	ScaleAblation (P1)	Full 582-dim vs. ablated 326-dim (without patchwise scale statistics) vs. 64-dim (statistics only) → ablation dataset triple for RQ1 and RQ3
3	Detector (P2)	Binary RF, 80/20 rebalanced, class_weight='balanced', RandomizedSearchCV tuning; Isolation Forest baseline → Trained Model 1; binary F1, precision/recall
4	Classifier (P2)	11-class RF on anomalous subset, RandomizedSearchCV tuning; stratified k-fold for small classes → Trained Model 2; per-class F1 table
5	SHAPExplainer (P2)	TreeExplainer on both models → SHAP arrays; top- <i>k</i> channel rankings per prediction with direction relative to normal-class mean
6	AlignmentTable (P3)	10 rows (jamming excluded); draft Wk5, validated Wk6. Two members independently map each fault using: (1) TelecomTS anomaly description, (2) GSMA KG keyword search, (3) primary 3GPP text confirmation; each row records document ID, release, clause, evidence span, relation type; disagreements adjudicated; ambiguous cases flagged. Wk1 go/no-go confirms KG granularity → validated static JSON lookup table
7	KGIndexer + RAGQueryConstructor (P3)	Embed/index 1,235 tickets and QnA reasoning traces (anomalous samples only); build SHAP-derived query template → ChromaDB collection; retrieved content; cosine similarity scores
8	LLMExplainer (P4)	Fuse retrieved content + alignment row; enforce JSON schema with constrained decoding; two-check clause validation; template fallback on parse failure → explanation JSON per sample; grounded/ungrounded flag
9	Evaluator (P4)	Track A (detection metrics), Track B (TeleQnA probe, G-Eval (4 dimensions; Sec. 7)), Track C (three-condition ablation) → results tables
10	Dashboard (P4)	Streamlit app: 5 panels (detection, SHAP chart, KPI signal plot, explanation, sources); deployed on HF Spaces → public URL; live demo for submission

7 Evaluation Plan

Detection: Precision/Recall/F1-macro benchmarked against Mantis ($P = R = F1 = 0.800$, TelecomTS Table 2) for Model 1, a reference point, not a like-for-like contest (Mantis uses a frozen backbone; our classifier trains from scratch on the same split). For Model 2, no published per-class F1 baseline exists; TelecomTS reports plain accuracy only (Toto 0.848, Mantis 0.590, GPT-4.1-with-context 0.227, Table 4); we report both per-class F1-macro and plain accuracy. Our per-class F1 table is the first such result on this task.

Jamming generalization probe: All 279 Jamming samples (the only real over-the-air anomaly) are completely excluded from 11-class RCA training and evaluated as a zero-exposure generalization test after the main evaluation; results reported separately.

Explanation: Standards-grounded explanation quality is evaluated using the G-Eval framework [11] with a project-defined telecom-specific rubric comprising four dimensions: (1) *Citation Validity*: the cited 3GPP TS reference is valid; (2) *Fault Specificity*: the correct fault type and key KPI signatures are identified; (3) *Actionability*: the recommended diagnostic step is supported by the cited clause and retrieved evidence; (4) *Causal Soundness*: the stated fault mechanism is consistent with the cited clause. Each dimension is scored on a 1–5 scale, and the overall score is the mean across the four dimensions. Evaluation is performed on 30 stratified explanations (3 per synthetic fault type; jamming excluded). The percentage of explanations citing a valid 3GPP clause is reported as an additional objective metric. TeleQnA is used as a secondary evaluation proxy.

Success criterion: We pre-register 70% as our working threshold for declaring the grounding contribution meaningful in this pilot. This is the team’s own hypothesis based on the general principle that retrieval-augmented context reduces hallucination in domain-specific generation; it is not a figure derived from any single prior publication, and the ungrounded baseline rate will be measured empirically and reported alongside results. At $n = 30$, the 95% bootstrap CI on a 70% rate is approximately ± 16 pp; results are therefore reported as directional pilot evidence with explicit confidence intervals, not as a powered hypothesis test.

Baselines: Mantis Table 2 for Model 1; Toto/Mantis Table 4 as RCA context for Model 2. **Track C** uses three controlled conditions (identical model, prompt schema, generation settings): (1) label only; (2) label + retrieved content; (3) full system with alignment table. Conditions 2 vs. 3 isolate the standards-grounding contribution specifically. **Ablations:** scale-preserving features on/off; alignment-table grounding on/off (core novelty test); full vs. reduced KPI-channel subsets.

Rigor: Stratified k-fold CV for smallest classes ($n = 48$, $n = 60$) with explicit CIs. Human evaluation by two independent reviewers recruited outside the development team, blinded to condition labels, with a 30-minute calibration session; disagreements adjudicated by a third reviewer; inter-rater agreement reported per dimension.

8 Expected Outcomes / Deliverables

- End-to-end working prototype (KPI → detection → classification → grounded explanation) via Streamlit, showing per-window detection output, SHAP importance plots, retrieved 3GPP/O-RAN citations, and diagnostic narratives; verified end-to-end at Week 11.
- Validated Fault-to-Standard Alignment Table (10 synthetic categories), publicly released, the first such mapping in the published literature; all 10 rows independently verified by two team members with zero unresolved disputes.
- Structured 30-sample G-Eval-annotated explanation set [11] scored across four project-defined dimensions.
- Experimental results: per-class F1 vs. baselines, three-condition explanation ablation, and three feature/grounding ablation studies; any reductions reported explicitly.
- Final report with explicit limitations section covering single testbed scope, 10/11 synthetic fault composition, synthetic RAG corpus, small-sample classes ($n = 48$, $n = 60$), and 30-sample pilot evaluation, and window-level train/test split with 75% adjacent-window overlap (recording-level split infeasible on this dataset; documented in Sec. 6.3).
- Modular, fully documented source code; each module independently runnable per the shared JSON schema.
- *Future work (out of scope)*: multi-cell correlation, cross-domain RAN/core/transport RCA, live production deployment.

9 Timeline (15 weeks: July 14 – October 31, 2026)

Ph.	Task	Wks
1	Literature review, dataset evaluation, alignment-table feasibility scoping, proposal drafting and submission	1–3
2	Architecture and pipeline design with evaluation harness; alignment-table draft against 3GPP source documents; inter-layer schema. <i>Checkpoint Wk5</i> : Alignment Table draft complete	4–5
3	Implementation (core): preprocessing, RF detection/classification + SHAP, RAG pipeline, LLM generation, full integration. Highest-risk phase, 6 weeks, 5 sub-systems (see Sec. 10). <i>Checkpoints</i> : Wk6: table validated; Wk7: RF trained; Wk8: Layer 1+2 integration (classifiers on held-out samples, SHAP generating, one sample through both models); Wk9: RAG returning results; Wk11: full end-to-end demo. <i>External MVP (Sep 12, approx. Wk9)</i> : Layer 1+2 complete with metrics logged; Layer 3 in active development targeting Wk11. <i>Contingency</i> : if Wk11 slips, Track C ablations reduce 3→2, prioritizing alignment-table grounding	6–11
4	Evaluation: detection metrics, G-Eval [11] human pilot evaluation(30 samples, two independent reviewers, begins in parallel with Wk11 to avoid bottlenecking ablation runs), groundedness and ablation studies	12–13
5	Analysis and writing: results, limitations, final report	14–15

10 Risks and Mitigation

Risk	Likelihood	Impact	Mitigation
GSMA KGs lack clause-level granularity	Medium	High	Wk1 go/no-go; fallback to keyword search in TS 38.300/38.104
Small fault classes (n=48, n=60) unstable	High (known)	Medium	Stratified k-fold CV; per-class CIs disclosed explicitly
LLM hallucinates citations	Medium	High	Two-check validation (format regex + alignment table) plus manual spot-check during G-Eval human pilot evaluation
SHAP integration delayed	Medium	Medium	RF chosen for native TreeExplainer; permutation importance fallback if Wk7 fails
RAG retrieval irrelevant	Medium	High	Precision@k at Wk9; prompt redesign if below threshold
Alignment Table mapping errors	Medium	High	All 10 rows independently verified by two members at Wk6
Groq API approval denied/delayed past Wk9	Medium	High	Governance request by Wk2; dev on HPC/Ollama so never blocked; fallback: classification-plus-template mode, offline human evaluation
HPC GPU unavailable Weeks 6–11	Medium	Medium	Prototype Layer 3 on CPU (Phi-3-mini/Llama 3.2 1B); request early; RF+SHAP CPU-only and unaffected
Over-investment in re-tuning past Wk5	Medium	Medium	Model-lock gate: if binary recall <0.50 and macro F1 <0.35 after RandomizedSearchCV, stop re-tuning and proceed to Layer 3 regardless; report the shortfall as a documented finding
External reviewers unavailable Wk12–13	Medium	Medium	P1/P2 (non-development roles) as fallback; conflict-of-interest disclosed; blinding still applied
Window-level split overlap (75% adjacent-window leakage)	Known	Low	Consistent with TelecomTS paper protocol; recording-level split infeasible on this dataset (zero anomalies in test under session assignment); reported explicitly in results as a limitation

11 Resources Required

Compute: RF and SHAP are CPU-only (no GPU required). LLM/RAG development and evaluation-phase inference: Queen’s HPC (Frontenac), A100 40GB preferred / V100 32GB minimum, 8+ cores, 64GB RAM, 100GB storage, Weeks 6 – 13. **Software:** scikit-learn, SHAP, sentence-transformers, ChromaDB, Streamlit, Ollama; all open-source, no paid licenses. **Datasets:** TelecomTS (MIT); GSMA KGs (CC BY-NC 4.0, reference only, not redistributed); TeleQnA (public); no access agreements required. **Advisor:** biweekly with Prof. Chen and Capgemini’s team.

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